

TRAPANI BIRGI, Italy

WMO: 164290

Lat: 37.911N

Lon: 12.488E

Elev: 24

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	39.3	41.3	32.1	26.5	50.7	35.4	30.3	48.3	33.1	55.3	30.2	56.0	3.6	120	0.443

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	16.0	93.2	72.3	89.8	73.9	87.4	74.9	81.6	86.0	80.0	84.7	78.6	83.5	13.8	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.5	158.3	85.3	78.7	149.0	83.9	77.0	140.8	82.5	45.3	86.0	43.6	84.8	42.0	83.8	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.5	25.4	22.0	35.3	100.4	2.2	3.5	33.7	102.9	32.4	104.9	31.1	106.9	29.5	109.5		
(5)			34.2	84.8	2.2	2.0	32.6	86.3	31.3	87.4	30.1	88.6	28.5	90.1	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	64.6	52.6	52.2	55.2	59.4	65.9	72.6	77.8	78.9	74.4	68.3	61.2	55.3	(6)	(7)
	DBStd	10.30	3.88	4.08	4.17	4.27	4.68	4.89	3.60	3.51	4.21	4.35	4.51	4.09	(7)	(8)
	HDD50	52	18	22	6	0	0	0	0	0	0	0	0	6	(8)	(9)
	HDD65	1713	383	358	305	177	42	1	0	0	0	17	129	301	(9)	(10)
	CDD50	5365	99	83	167	283	492	679	861	894	733	566	337	171	(10)	(11)
	CDD65	1549	0	0	1	9	68	230	396	429	283	118	15	0	(11)	(12)
	CDH74	13247	0	0	6	51	430	1761	4034	4551	1999	402	13	0	(12)	(13)
	CDH80	4169	0	0	1	9	110	527	1379	1659	446	38	0	0	(13)	(14)
(14) Wind	WSAvg	8.9	9.7	9.8	9.9	9.5	8.6	7.7	7.6	7.4	8.2	8.1	10.1	10.0	(14)	(15)
(15) Precipitation	PrecAvg	18.8	2.4	2.0	1.7	1.5	0.7	0.2	0.1	0.4	1.8	2.6	2.8	2.7	(15)	(16)
	PrecMax	34.4	6.7	6.3	5.3	6.2	2.7	0.9	1.3	3.1	6.0	7.0	11.8	7.6	(16)	(17)
	PrecMin	7.9	0.4	0.0	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	(17)	(18)
	PrecStd	5.9	1.4	1.3	1.3	1.2	0.6	0.2	0.2	0.8	1.5	1.8	2.1	1.5	(18)	(19)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.6	66.5	73.0	79.3	89.3	96.5	97.2	98.7	91.5	83.9	75.6	68.2	(19)	(20)
		MCWB	59.4	59.0	59.9	59.7	65.4	70.5	71.5	74.1	73.2	70.9	68.0	63.2	(20)	(21)
	2%	DB	62.6	62.8	67.9	74.1	82.4	89.9	92.3	93.0	87.5	80.5	73.2	65.3	(21)	(22)
		MCWB	58.3	57.8	58.8	60.3	63.9	69.2	71.9	74.9	74.7	71.4	66.2	61.0	(22)	(23)
	5%	DB	60.8	60.9	64.6	70.4	78.6	85.7	89.1	89.6	84.4	78.1	70.6	63.3	(23)	(24)
		MCWB	57.0	56.7	58.7	59.9	63.9	70.0	74.1	76.4	74.9	70.8	64.7	59.3	(24)	(25)
	10%	DB	59.2	59.3	62.6	67.6	75.1	82.1	86.2	86.9	82.3	76.0	68.4	62.0	(25)	(26)
		MCWB	55.6	55.2	58.1	59.7	64.6	70.9	75.5	77.2	74.2	69.5	63.2	58.3	(26)	(27)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.6	60.6	63.3	66.3	71.5	79.7	83.3	84.6	80.8	75.8	70.8	63.9	(27)	(28)
		MCDB	63.2	64.4	67.4	71.5	76.6	84.8	86.8	89.0	85.2	79.8	74.3	66.7	(28)	(29)
	2%	WB	58.9	58.7	61.0	63.6	69.4	76.9	81.3	82.3	78.5	73.6	67.5	61.8	(29)	(30)
		MCDB	61.6	61.9	64.8	69.0	75.1	81.4	85.4	86.9	83.1	77.9	71.2	64.3	(30)	(31)
	5%	WB	57.5	57.3	59.5	62.0	68.0	75.1	79.8	80.6	76.9	72.0	65.7	60.2	(31)	(32)
		MCDB	60.2	60.3	63.3	66.8	73.6	80.3	84.2	85.4	82.0	76.3	69.4	62.7	(32)	(33)
	10%	WB	56.1	55.8	58.2	60.7	66.4	73.4	78.3	79.1	75.3	70.5	64.1	58.8	(33)	(34)
		MCDB	58.8	58.7	61.9	65.4	72.1	78.9	82.9	84.2	80.2	74.8	67.8	61.4	(34)	(35)
(35) Mean Daily Temperature Range	5% DB	MDBR	11.9	12.2	13.5	14.8	15.7	16.5	16.2	16.0	14.2	13.8	12.1	11.2	(35)	(36)
		MCDBR	12.5	13.7	16.6	20.1	22.0	23.1	21.9	20.4	16.8	15.1	13.4	12.2	(36)	(37)
	5% WB	MCWBR	10.1	10.9	11.9	11.7	11.6	12.9	14.1	13.8	10.8	10.1	9.7	9.6	(37)	(38)
		MCWBR	11.6	12.7	15.1	16.4	18.1	17.6	16.6	16.7	15.0	14.2	12.7	11.0	(38)	(39)
(39) Clear-Sky Solar Irradiance	taub	taub	0.349	0.370	0.412	0.451	0.457	0.431	0.438	0.435	0.440	0.416	0.383	0.356	(39)	(40)
		taud	2.452	2.373	2.245	2.152	2.179	2.299	2.292	2.305	2.290	2.330	2.370	2.425	(40)	(41)
	Edn at Noon	Edn at Noon	262	270	268	264	264	270	267	266	256	250	245	249	(41)	(42)
		Edh at Noon	27	33	41	47	47	42	42	40	38	34	29	26	(42)	(43)
(43) All-Sky Solar Radiation	RadAvg	RadAvg	709	980	1405	1817	2162	2423	2446	2149	1610	1172	809	634	(43)	(44)
	RadStd	RadStd	52	90	111	114	110	69	52	83	71	72	49	53	(44)	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	-0.81	-0.99	N/A	N/A	N/A	N/A

Nomenclature: See separate page